



Gubbala Kundana Satya Deepak

Roll No: 220101043

B.Tech - Computer Science and Engineering

Indian Institute Of Technology, Guwahati

+91-9182748959

gksdeepak146@gmail.com

d.gubbala@iitg.ac.in

Github | linkedin

EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech	Indian Institute of Technology, Guwahati	7.55	2022-Present
Senior Secondary	Board of Intermediate Education, Andhra Pradesh	97.4%	2022
Secondary	Board of Secondary Education, Andhra Pradesh	100%	2020

EXPERIENCE

- Samsung Research Institute Bangalore (SRIB)** May - Jul. 2025
Software Developer Intern Bengaluru, India
 - Developed a **GenAI** application to automate Root Cause Analysis (RCA), reducing manual debugging time by an estimated **40%** through orchestration of an end-to-end **RAG** pipeline with **LangChain** and **Amazon Bedrock(AWS)**.
 - Automated data ingestion by integrating **Jira** for incident tickets and **Grafana Loki** for log retrieval.
 - Built a self-improving **knowledge base** with a **Weaviate** vector DB and a Streamlit-based **Human-in-the-Loop (HITL)** interface, leveraging **user-verified RCAs** to progressively improve future analysis accuracy.

PROJECTS

- CodeFlow** May - Jun. 2024
Coding Club, IIT Guwahati Github
 - Developed a **MERN**-based web application to assist coders in enhancing their Competitive Programming skills.
 - Implemented features like secure **user authentication**, email verification with **Nodemailer**, problem browsing, in-browser **code editor**, efficient **bookmarking** for quick access, and active engagement through **comments**.
 - Integrated the **Clist API** to aggregate and display real-time contest schedules from **15+** coding platforms.
- QoR Predictor for AIGs** Jan. - Apr. 2025
Course Project, ML for EDA Github
 - Developed an ML model combining **GNN** and **LSTM** to predict circuit area after logic synthesis optimizations.
 - Built an **RL agent** using policy gradient methods to automatically generate efficient synthesis recipes for **AIGs**.
 - Processed **circuits as graphs** and synthesis **sequences** as temporal data using **PyTorch** and **PyTorch Geometric**.
 - Achieved a **2-6%** prediction error and **accelerated** recipe discovery by reducing design space exploration.
- Campus Ambassador Portal** Apr. - Jun. 2024
Spirit Web Operations Team Github
 - Built the **official Spirit 2024** portal, a **full-stack web** application that streamlined registration for over **2,000** users.
 - Implemented a secure user authentication system with **Argon2** password hashing and **session management**.
 - Developed and integrated a **live scoreboard** to display **real-time** competition results and participant rankings.
 - Architected the backend using **Node.js** for robust server-side logic and **MySQL** for persistent data storage.
- Other Projects**
 - Bachelor's Thesis Project:** Researching Vision Language Models (**VLMs**) for visual search applications by applying prompt engineering and **fine-tuning** techniques like visual prompt tuning (**VPT**).
 - Antahsagari, Underwater ROV:** Developed a Python-based **image processing pipeline** using a **Super-Resolution** model and OpenCV to **enhance and denoise** underwater ROV footage for visual analysis.

TECHNICAL SKILLS

Programming: C, C++, Python.

Database: MySQL, MongoDB*.

Operating Systems: Windows, Linux*

Web Technologies: Node.js, React.js, Streamlit.

Tools/Frameworks: TensorFlow, OpenCV, LangChain*.

Miscellaneous: Git, GitHub, Docker*.

* Elementary proficiency

ACHIEVEMENTS

- Specialist**, in Codeforces rating **1400+**, Global Rank **1326** among **26k+** participants (Round 1051). handle
- CodeChef**, Global Rank **233** (starters 201), Global Rank **240** (starters 199) 2025
- AMUROV Competition**, led the Antahsagari robotics team to the final offline round. 2024
- JEE Advanced**, Secured **All India Rank 1844** among **1,50,000** students. 2022
- JEE Mains**, Secured **All India Rank 1732 (99.81 percentile)** among **1 million** students. 2022
- TS EAMCET**, Secured a rank of **279** among **1,70,000** students. 2022

KEY COURSES TAKEN

- Computer Science:** Data Structures and Algorithms, Software Engineering, Randomized Algorithms, Speech Processing, High Performance Computing, Operating Systems, Computer Networks.
- Data Science:** Machine Learning, Hardware-aware Deep Learning, Database Management Systems, ML for EDA.
- Mathematics:** Probability Theory and Random Processes, Linear Algebra, Discrete Mathematics, Optimization.

POSITIONS OF RESPONSIBILITY

- WebOps Core Team Member**, Spirit, IIT Guwahati Mar. 2024 - Apr. 2025
- Project Lead**, 4i labs, IIT Guwahati Apr. 2024 - Apr. 2025
 - Led a multidisciplinary team in robotics research and development, with a focus on ROV mechanical design, control systems, and underwater image processing.